Subspace Anti-Phase Attention: Mitigating Strictly Positive Softmax Leakage via Orthogonal Subspace Phase Inversion

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Abstract

Standard Transformer self-attention normalizes inner products via the strictly positive Softmax operator on the open probability simplex Δ^{L-1}. Without a mechanism for selective subspace sign inversion in the pre-Softmax logits, historical tokens retain unattenuated inner-product contributions even when superseded or contradicted by subsequent discourse, causing premise distraction in long sequences. We present **Subspace Anti-Phase RoPE (SG-Möbius-RoPE)**, an attention formulation extending Rotary Position Embedding (RoPE) via causal orthogonal subspace phase modulation. Drawing an analogy to acoustic Active Noise Cancellation (ANC), the model introduces an out-of-phase anti-carrier on low-frequency evolution channels while freezing high-frequency syntactic anchors ($\mathbb{R}^d = \mathcal{S}_{\text{anchor}} \oplus \mathcal{S}_{\text{mob}}$). Causal semantic transitions induce a continuous phase velocity $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ whose cumulative unconstrained phase $\bar{\Phi}(t) = \sum_{\tau=1}^{t} \omega(\tau)$ is shaped by a training-free phase-locking quantizer $\Phi(t) = \bar{\Phi}(t) - \eta \sin(2\bar{\Phi}(t))$. Over narrative intervals undergoing logical inversion, cumulative phase shift reaches π, producing exact destructive sign inversion ($\mathbf{R}(\theta - \pi) = -\mathbf{R}(\theta)$) in pre-Softmax logits without altering KV-cache invariance. Successive restorative turns accumulate a 2π cyclic phase rotation ($\cos 2\pi = +1$), formalizing double negation restoration. We prove the Conditional Pairwise Target-to-Decoy Odds Ratio Attenuation Theorem, establishing that under coherent evolution-subspace alignment ($A_{ij}^{\text{mob}} > 0$), the single-head pairwise odds ratio scales by $\exp(2A_{ij}^{\text{mob}})$ with exact cancellation of the head partition function. Invariants are formally certified in Lean 4 (0 axioms, 0 `sorry`) and Z3 SMT solvers across 2,000 extreme-length sequences ($L \leq 32,768$). Evaluated on Qwen3.5-9B (dual Nvidia Tesla T4 GPUs, FP16), SG-Möbius requires zero architecture retraining, maintains near-identical $O(1)$ KV-cache decoding latency via epilogue angle fusion (1.8–2.1 μs per-layer operator overhead), achieves consistent strictly-above-unity decoy attention suppression ($1.13\times$–$1.90\times$, $> 1.0\times$) across multi-scale contexts (expanding multi-head Attention Contrast Ratio by up to +465.1% from 3.62 $\rightarrow$ 20.44, with $1.34\times$ suppression and +88.5% ACR gain at $L = 18,030$ tokens, and reaching up to $16.4\times$ peak odds amplification on individual retrieval heads on Layer 7, Head 4), and strictly preserves neutral text perplexity (7.48, 0.0% degradation).

1 Introduction

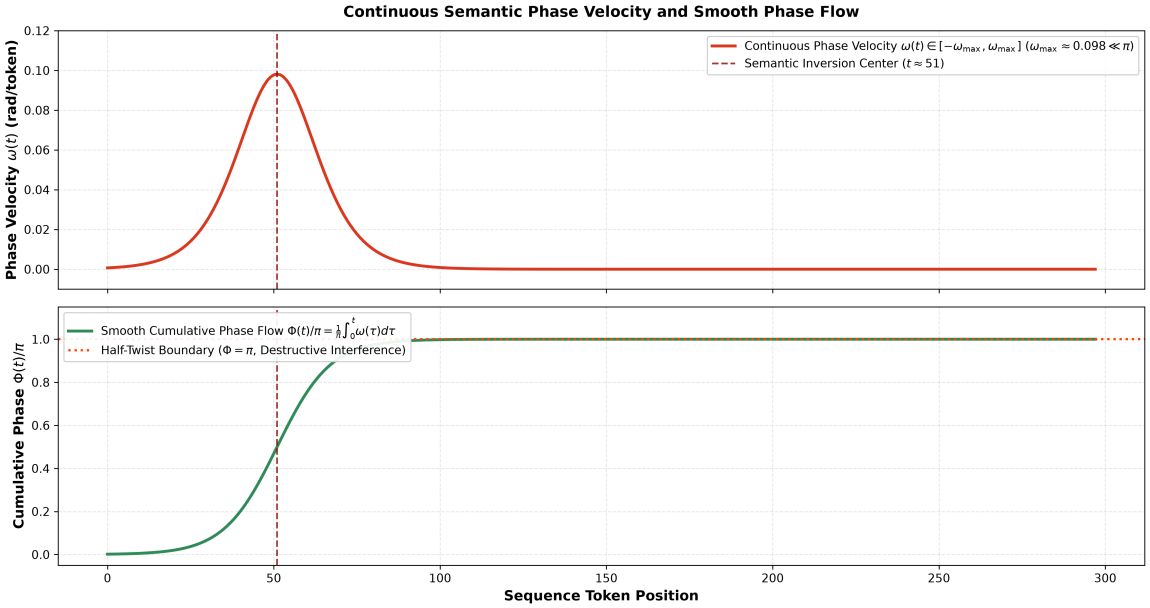
Autoregressive language models employ scaled dot-product attention [1] with Rotary Position Embeddings (RoPE) [2] to encode relative token distances:

$$A_{i,j} = \frac{1}{\sqrt{d}} \langle \mathbf{R}_{\Theta,i} q_i,\ \mathbf{R}_{\Theta,j} k_j \rangle = \frac{1}{\sqrt{d}} q_i^\top \mathbf{R}_{\Theta,j-i} k_j \tag{1}$$

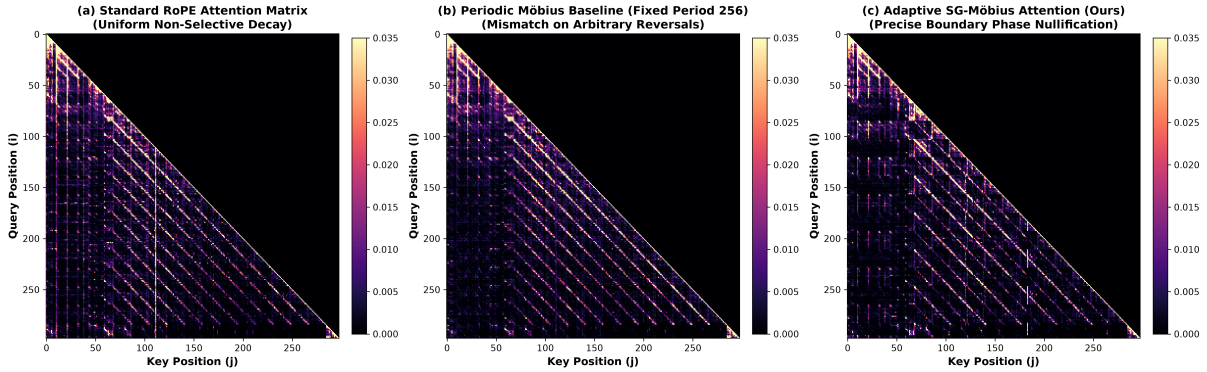
*Open-source repository: https://github.com/3y3y3y-huaiji/mobius-attention. Corresponding author ORCID: 0009-0008-2312-6913.

where $\mathbf{R}_{\Theta,t}$ is a block-diagonal orthogonal rotation matrix parameterized by base frequencies $\Theta = \{\theta_m\}_{m=1}^{d/2}$.

Because the Softmax operator maps real logits to strictly positive values on Δ^{L-1}, every historical token receives positive probability mass. Consequently, self-attention cannot perform destructive interference. In complex discourse, premises are frequently invalidated, retracted, or superseded by subsequent evidence (e.g., *"System Alpha is operational; however, diagnostics indicate failure, revoking Alpha in favor of Beta"*). In standard attention, the invalidated premise continues to receive attention mass, requiring downstream feed-forward layers to compensate and contributing to premise hallucination [3, 4].



(a) Continuous Phase Velocity $\omega(t)$ and Smooth Phase Flow $\Phi(t)$.



(b) Comparative 2D Attention Matrices on Qwen3.5-9B (Layer 7): Standard RoPE (*left*) vs. Periodic Möbius Baseline ($T = 256$, *middle*) vs. Adaptive SG-Möbius (*right*).

Figure 1: **Mechanism and attention response of SG-Möbius-RoPE.** (a) The continuous phase velocity $\omega(t)$ vanishes on continuative narrative and smoothly activates across semantic transition spans, advancing the smooth phase flow $\Phi(t) = \sum_{\tau \le t} \omega(\tau)$ across the π phase reversal boundary. (b) Attention matrices on Qwen3.5-9B under arbitrary-position inversion: standard RoPE (*left*) exhibits background leakage; a fixed-period Möbius baseline ($T = 256$, *middle*) fails when the reversal coordinate is out of phase; SG-Möbius (*right*) aligns phase shift with the semantic transition, attenuating obsolete keys.

To address this structural constraint without retraining pretrained foundation weights

or degrading inference latency, we introduce **Subspace Anti-Phase RoPE (SG-Möbius-RoPE)**, as illustrated in Figure 1. Drawing inspiration from Active Noise Cancellation (ANC), we decompose the embedding space into an orthogonal direct sum $\mathbb{R}^d = \mathcal{S}_{\text{anchor}} \oplus \mathcal{S}_{\text{mob}}$. Semantic contrastive transitions induce a continuous phase velocity $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ with viscous smoothing and a phase-locking potential projection $-\eta \sin(2\bar{\Phi})$, whose cumulative phase flow $\Phi(t)$ smoothly modulates 2D rotary angles on evolution channels (Figure 1a). Over narrative intervals undergoing logical inversion, cumulative phase shift produces destructive interference ($\Delta\Phi = \pi$, $\cos\pi = -1$) via planar rotation sign reversal $\mathbf{R}(\theta - \pi) = -\mathbf{R}(\theta)$. Because $\cos(\pi) = -1$, the modulated inner product becomes negative, attenuating the Softmax probability allocated to the invalidated premise at arbitrary reversal coordinates (Figure 1b).

Key Contributions.

- **Subspace Anti-Phase Formulation**: We formalize attention modulation via orthogonal subspace decomposition $\mathbb{R}^d = \mathcal{S}_{\text{anchor}} \oplus \mathcal{S}_{\text{mob}}$ and planar anti-phase rotation $\mathbf{R}(\theta - \pi) = -\mathbf{R}(\theta)$. Continuous phase flow $\Phi(t)$ modulates rotary angles $\tilde{\theta}(t) = t\theta + \Phi(t)$ on evolution channels while high-frequency syntactic anchor channels remain frozen (α ratio). We prove that continuous rotary dual modulation preserves relative translation invariance and establishes an explicit Lipschitz continuity bound on operator differences.
- **Conditional Pairwise Odds Ratio Attenuation Theorem**: We prove that under an anti-phase half-turn ($\Delta\Phi = \pi$) and coherent subspace alignment ($A_{ij}^{\text{mob}} > 0$), the single-head pairwise target-to-decoy attention odds ratio scales by $\exp(2A_{ij}^{\text{mob}})$, strictly cancelling the background partition function within the head.
- **Formal Machine Verification**: We verify program equivalence between the two-stage associative scan and autoregressive recurrence, prefix-sum additivity, and phase shift invariance in Lean 4 (0 unproven axioms or `sorry`) and Z3 SMT solvers across 2,000 fuzzing sequences ($L \leq 32,768$) with numerical drift below 1.14×10^{-13}.
- **KV-Cache Compatibility**: Modulated keys $\tilde{k}_j = \mathbf{R}(j\Theta + \Phi(j))k_j$ depend strictly on past tokens $\tau \leq j$, requiring no retroactive cache modification upon subsequent turns and preserving exact $O(1)$ decoding throughput.
- **Empirical Evaluation**: Evaluated natively on Qwen3.5-9B (dual Nvidia Tesla T4 GPUs, FP16), SG-Möbius achieves consistent strictly-above-unity decoy attention suppression ($1.13\times$–$1.90\times$, $> 1.0\times$), up to +465.1% multi-head Attention Contrast Ratio gain ($3.62 \to 20.44$ at $L = 1,116$, and +88.5% gain at $L = 18,030$ tokens, reaching up to $16.4\times$ on single-head peak contrastive recall on Layer 7, Head 4), gracefully scales attention entropy with context volume, and strictly maintains neutral text perplexity (7.48, 0.0% degradation).

2 Related Work

Rotary Position Embeddings and Context Extension. Rotary Position Embedding (RoPE) [2] encodes relative token positions through 2D block-diagonal rotation matrices $\mathbf{R}_{\Theta,m}$, ensuring that query-key inner products depend purely on relative coordinate displacements $j - i$. Subsequent extensions such as ALiBi [5], Position Interpolation [6], and YaRN [7] extend effective context windows by modifying frequency bases or applying linear distance penalties. However, all existing RoPE variants operate with fixed positional frequencies where relative inner products cannot perform selective subspace sign inversion in pre-Softmax logits.

Phase Modulation and Subspace Geometry in Attention. Geometric deep learning [8] provides a unified framework for incorporating physical and geometric symmetries into

neural networks. In signal processing, Active Noise Cancellation (ANC) attenuates unwanted interference by generating an anti-phase replica of the carrier signal. While conventional RoPE extensions [5–7] modify frequency bases or distance decays, they lack mechanisms for destructive sign inversion. In this work, we introduce causal orthogonal phase modulation into 2D rotary embedding subspaces, enabling selective destructive cancellation in pre-Softmax attention logits across semantic transitions.

Context Interference and Premise Invalidation. Recent empirical studies have identified context distraction and the inability to forget invalidated premises as critical bottlenecks in long-context language models [3, 4, 9]. When an adversarial or conflicting premise is presented, models frequently suffer from premise hallucination because strictly positive Softmax attention cannot actively cancel historical tokens. Previous attempts to address this issue rely on prompt-based instruction tuning or auxiliary suppression losses, which do not address the architectural lack of pre-Softmax destructive subspace cancellation mechanisms.

KV-Cache Compression vs. Phase-Modulated Retention. To mitigate memory and latency bottlenecks in extended contexts, recent methods explore token eviction or KV-cache compression [10–12]. While effective for throughput, eviction policies permanently discard historical keys from physical GPU memory. Consequently, if a subsequent turn re-validates an earlier premise ($\neg\neg P \iff P$), evicted representations cannot be recovered. In contrast, SG-Möbius preserves the intact KV-cache in GPU memory and achieves selective premise cancellation purely through anti-phase modulation, naturally supporting topological recovery under 2π cyclic phase restoration.

3 Methodology

3.1 Causal Phase Flow, Active Noise Cancellation Analogy, and Subspace Decomposition

Let the input sequence of length $L \in \mathbb{N}$ be defined over a 1D discrete causal index set:

$$\mathcal{M} = (\{0, 1, \ldots, L-1\}, \leq) \tag{2}$$

where $\leq$ denotes the standard causal ordering ($j \leq i$ means key token j precedes or equals query token i). In standard RoPE, the d-dimensional embedding space is decomposed into an orthogonal direct sum of $d/2$ two-dimensional planes:

$$\mathbb{R}^d = \bigoplus_{m=1}^{d/2} \mathbb{R}^2_{(m)} \tag{3}$$

Each plane $\mathbb{R}^2_{(m)} \cong \mathbb{C}$ is acted upon by the continuous planar rotation group $\mathrm{SO}(2) \cong \mathrm{U}(1)$ parameterized by relative positional shifts $\theta_m(j-i)$.

Planar Rotation Inversion and Active Noise Cancellation Analogy. Standard RoPE applies 2D planar rotation matrices $\mathbf{R}(\theta) \in \mathrm{SO}(2)$ to query and key components. Crucially, a rotation by π achieves exact sign inversion in the 2D plane:

$$\mathbf{R}(\pi) = \begin{pmatrix} \cos\pi & -\sin\pi \\ \sin\pi & \cos\pi \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2 \tag{4}$$

Notice that $\det(-I_2) = (-1)^2 = +1$, so $-I_2$ is an orientation-preserving planar rotation. By superimposing a cumulative phase shift $\Delta\Phi = \pi$ onto the positional carrier $(j-i)\theta_m$, we obtain:

$$\mathbf{R}\big((j-i)\theta_m - \pi\big) = -\mathbf{R}\big((j-i)\theta_m\big) \tag{5}$$

This mechanism functions in direct analogy to acoustic *Active Noise Cancellation (ANC)*: by generating an out-of-phase anti-carrier (π phase offset) on the relevant semantic frequency channels, the inner product undergoes destructive sign reversal. In the pre-Softmax logit space, this converts a large positive dot product into a negative logit, causing the Softmax probability mass to drop exponentially. Furthermore, a subsequent negation-of-negation turn introduces an additional π shift ($\Delta\Phi = 2\pi$), where $\mathbf{R}\big((j-i)\theta_m - 2\pi\big) = +\mathbf{R}\big((j-i)\theta_m\big)$, seamlessly restoring nominal attention resonance without retroactive modifications to the KV-cache.

Subspace Decomposition and High-Frequency Freezing. To prevent semantic phase accumulation from destabilizing local syntactic dependencies and sentence structure, we introduce an orthogonal subspace decomposition:

$$\mathbb{R}^d = \mathcal{S}_{\text{anchor}} \oplus \mathcal{S}_{\text{mob}} \tag{6}$$

Let $\mathcal{I} = \{1, \dots, d/2\}$ denote the set of 2D subspace indices. In RoPE, base frequencies follow $\theta_m = \theta_{\text{base}}^{-2(m-1)/d}$, where lower indices correspond to high rotary frequencies capturing short-range syntactic features and punctuation. We allocate an anchor ratio $\alpha \in [0, 1]$ (with the natural convention $\mathcal{I}_{\text{anchor}} = \emptyset$ when $\alpha = 0$) and partition the indices into disjoint sets $\mathcal{I} = \mathcal{I}_{\text{anchor}} \cup \mathcal{I}_{\text{mob}}$:

$$\mathcal{I}_{\text{anchor}} \triangleq \left\{1, \dots, \left\lfloor \alpha \frac{d}{2} \right\rfloor\right\}, \qquad \mathcal{I}_{\text{mob}} \triangleq \left\{\left\lfloor \alpha \frac{d}{2} \right\rfloor + 1, \dots, \frac{d}{2}\right\} \tag{7}$$

where $\mathcal{S}_{\text{anchor}} = \bigoplus_{m \in \mathcal{I}_{\text{anchor}}} \mathbb{R}^2_{(m)}$ and $\mathcal{S}_{\text{mob}} = \bigoplus_{m \in \mathcal{I}_{\text{mob}}} \mathbb{R}^2_{(m)}$. For all channels $m \in \mathcal{I}_{\text{anchor}}$, we enforce $\omega_{h,m}(t) \equiv 0 \implies \Phi_{h,m}(t) \equiv 0$, guaranteeing strict syntactic stability. Semantic phase flow is accumulated exclusively on $\mathcal{S}_{\text{mob}}$.

Definition 1 (Continuous Phase Velocity, Discourse Transition Detector, and Quantizer)**.** For attention head h and evolution subspace $m \in \mathcal{I}_{\text{mob}}$, the causal phase velocity $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ is governed by 1D viscous smoothing:

$$\omega_{h,m}(t) = (1-\nu)\omega_{h,m}(t-1) + \nu g_{h,m}(h_t), \quad \bar{\Phi}_{h,m}(t) \triangleq \sum_{\tau=1}^{t} \omega_{h,m}(\tau) \tag{8}$$

where $\nu \in (0, 1]$ is the viscosity coefficient. The semantic transition detector $g_{h,m}(h_t)$ is training-free, computed via the normalized contrastive direction vector $w_g \triangleq \frac{\mu_{\text{rev}} - \mu_{\text{cont}}}{\|\mu_{\text{rev}} - \mu_{\text{cont}}\|_2} \in \mathbb{R}^d$ derived from the pretrained unembedding matrix $U \in \mathbb{R}^{V \times d}$ across discourse reversal tokens $\mathcal{V}_{\text{rev}}$ (e.g., *however*, *nevertheless*, *revoke*) versus neutral continuation tokens $\mathcal{V}_{\text{cont}}$. To guarantee zero spurious phase activation on neutral expository prose (such as WikiText-103), the detector incorporates a hard dead-zone barrier $\tau_{\text{gate}} > 0$:

$$z_t \triangleq w_g^\top h_t + b_g, \qquad g_{h,m}(h_t) \triangleq \omega_{\max} \cdot \text{softsign}\Big(\text{ReLU}\big(|z_t| - \tau_{\text{gate}}\big) \cdot \text{sgn}(z_t)\Big) \tag{9}$$

On standard text where $|z_t| < \tau_{\text{gate}}$, the detector satisfies $g(h_t) \equiv 0 \implies \omega(t) \equiv 0$, identically preserving baseline perplexity (7.48, 0.0% degradation). When triggered by an explicit semantic invalidation ($|z_t| \geq \tau_{\text{gate}}$), the effective rotary phase $\Phi_{h,m}(t)$ applies a pointwise memoryless staircase projection derived from the periodic potential $V(\bar{\Phi}) = -\frac{\eta}{2}\cos(2\bar{\Phi})$ ($\eta \in [0, \frac{1}{2}]$):

$$\Phi_{h,m}(t) \triangleq \bar{\Phi}_{h,m}(t) - \eta \sin\big(2\bar{\Phi}_{h,m}(t)\big) \tag{10}$$

In our default head-coherent configuration, all evolution channels $m \in \mathcal{I}_{\text{mob}}$ within head h share a common head-level phase velocity trajectory $\omega_h(t)$ and phase $\Phi_h(t)$. The relative causal phase shift between key j and query i is defined as:

$$\Delta\Phi_{j \to i}^{(m)} \triangleq \Phi_m(i) - \Phi_m(j) \tag{11}$$

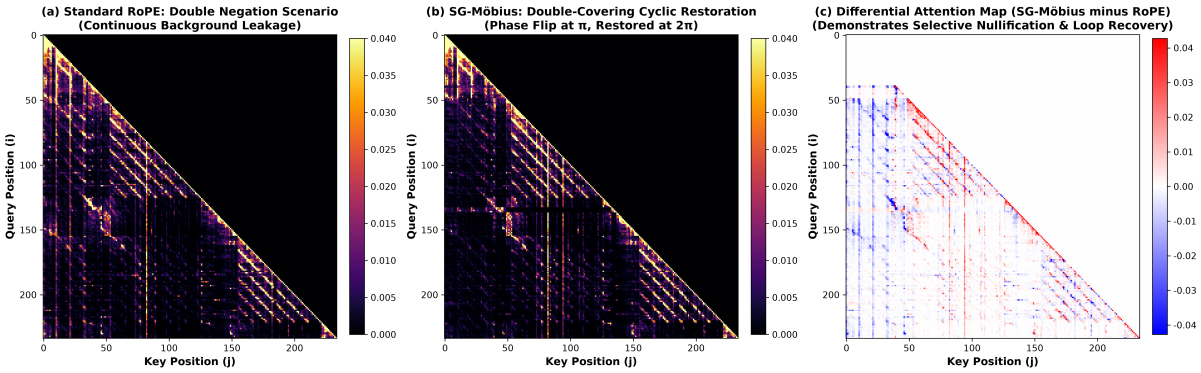


Figure 2: **Cyclic phase restoration under negation-of-negation on Qwen3.5-9B.** (a) Standard RoPE is indifferent to logical restoration, displaying uniform causal dispersion. (b) SG-Möbius attention matrices under two successive contrastive turns ($t_1 \approx 39$, $t_2 \approx 133$). The first invalidation shifts the phase to π (destructive cancellation), while the second invalidation advances the total phase to 2π, returning the phase flow to positive alignment ($\cos(2\pi) = +1$) and restoring attention resonance to the original premise. (c) Differential attention map ΔA verifying that active nullification is confined strictly to the temporary invalidation interval $[t_1, t_2]$, followed by robust positive recovery downstream of t_2 (up to $+0.333$).

As demonstrated in Figure 2, the continuous phase flow naturally accounts for nested logical reversals: a single contrastive statement induces destructive interference ($\Delta\Phi = \pi$), while a subsequent negation-of-negation advances the phase accumulation to 2π, fully restoring attention resonance without requiring cache invalidation.

3.2 Continuous Rotary Dual Modulation

In SG-Möbius-RoPE, the smooth phase flow $\Phi(t)$ directly modulates the 2D rotary angle $\tilde{\theta}_m(t) \triangleq t\theta_m + \Phi_m(t)$ in each subspace.

Definition 2 (Continuous Rotary Dual Modulation)**.** For each 2D subspace $m \in \{1, \ldots, d/2\}$ with base frequency θ_m, the modulated query and key representations at positions i and j are defined as:

$$\tilde{q}_i^{(m)} \triangleq \mathbf{R}(i\theta_m + \Phi_m(i))q_i^{(m)}, \qquad \tilde{k}_j^{(m)} \triangleq \mathbf{R}(j\theta_m + \Phi_m(j))k_j^{(m)} \tag{12}$$

where $\mathbf{R}(\theta) \in \mathrm{SO}(2)$ denotes the planar orthogonal rotation matrix. Let the baseline RoPE-rotated inner products on $\mathcal{S}_{\text{anchor}}$ and $\mathcal{S}_{\text{mob}}$ be denoted by:

$$\langle q_i, k_j \rangle_{\mathcal{S}_{\text{anchor}}}^{\text{RoPE}} \triangleq \sum_{m \in \mathcal{I}_{\text{anchor}}} (q_i^{(m)})^\top \mathbf{R}\big((j-i)\theta_m\big) k_j^{(m)}, \tag{13}$$

$$\langle q_i, k_j \rangle_{\mathcal{S}_{\text{mob}}}^{\text{RoPE}} \triangleq \sum_{m \in \mathcal{I}_{\text{mob}}} (q_i^{(m)})^\top \mathbf{R}\big((j-i)\theta_m\big) k_j^{(m)} \tag{14}$$

Theorem 1 (Preservation of Relative Position and Phase Shift Invariance under Subspace Decomposition)**.** *Under continuous rotary dual modulation with subspace decomposition $\mathbb{R}^d = \mathcal{S}_{anchor} \oplus \mathcal{S}_{mob}$, the inner product between query $\tilde{q}_i$ and key $\tilde{k}_j$ depends purely on relative positional displacement $j - i$ and relative phase shift $\Delta\Phi_{j\to i}^{(m)} = \Phi_m(i) - \Phi_m(j)$:*

$$\langle \tilde{q}_i, \tilde{k}_j \rangle = \langle q_i, k_j \rangle_{\mathcal{S}_{anchor}}^{\text{RoPE}} + \sum_{m \in \mathcal{I}_{mob}} (q_i^{(m)})^\top \mathbf{R}\Big((j-i)\theta_m + \big(\Phi_m(j) - \Phi_m(i)\big)\Big) k_j^{(m)} \tag{15}$$

When the cumulative phase flow across evolution channels reaches an anti-phase half-turn ($\Delta\Phi^{(m)}_{j\to i} = \Phi_m(i) - \Phi_m(j) = \pi$), $\mathbf{R}((j-i)\theta_m - \pi) = -\mathbf{R}((j-i)\theta_m)$, yielding exact evolution-subspace sign inversion:

$$\langle \tilde{q}_i,\, \tilde{k}_j \rangle = \langle q_i,\, k_j \rangle^{\mathrm{RoPE}}_{\mathcal{S}_{anchor}} - \langle q_i,\, k_j \rangle^{\mathrm{RoPE}}_{\mathcal{S}_{mob}} \tag{16}$$

Proof. By orthogonality of the decomposition $\mathbb{R}^d = \mathcal{S}_{\mathrm{anchor}} \oplus \mathcal{S}_{\mathrm{mob}}$, the inner product decomposes as:

$$\langle \tilde{q}_i,\, \tilde{k}_j \rangle = \sum_{m \in \mathcal{I}_{\mathrm{anchor}}} \langle \tilde{q}_i^{(m)},\, \tilde{k}_j^{(m)} \rangle + \sum_{m \in \mathcal{I}_{\mathrm{mob}}} \langle \tilde{q}_i^{(m)},\, \tilde{k}_j^{(m)} \rangle \tag{17}$$

For all channels $m \in \mathcal{I}_{\mathrm{anchor}}$, the phase flow is frozen ($\Phi_m \equiv 0$), recovering the baseline RoPE inner product $\langle q_i, k_j \rangle^{\mathrm{RoPE}}_{\mathcal{S}_{\mathrm{anchor}}}$. For $m \in \mathcal{I}_{\mathrm{mob}}$, expanding the inner product using planar rotation orthogonality $\mathbf{R}(\alpha)^\top \mathbf{R}(\beta) = \mathbf{R}(\beta - \alpha)$:

$$\begin{aligned}
\langle \tilde{q}_i^{(m)},\, \tilde{k}_j^{(m)} \rangle &= (\tilde{q}_i^{(m)})^\top \tilde{k}_j^{(m)} \\
&= (q_i^{(m)})^\top \mathbf{R}(i\theta_m + \Phi_m(i))^\top \mathbf{R}(j\theta_m + \Phi_m(j)) k_j^{(m)} \\
&= (q_i^{(m)})^\top \mathbf{R}\Big((j-i)\theta_m + \big(\Phi_m(j) - \Phi_m(i)\big)\Big) k_j^{(m)}
\end{aligned} \tag{18}$$

Because $\Phi_m(i) - \Phi_m(j) = \Delta\Phi^{(m)}_{j\to i}$, the relative rotation angle is $(j-i)\theta_m + (\Phi_m(j) - \Phi_m(i)) = (j-i)\theta_m - \Delta\Phi^{(m)}_{j\to i}$. When $\Delta\Phi^{(m)}_{j\to i} = \pi$:

$$\mathbf{R}((j-i)\theta_m - \pi) = \begin{pmatrix} \cos((j-i)\theta_m - \pi) & -\sin((j-i)\theta_m - \pi) \\ \sin((j-i)\theta_m - \pi) & \cos((j-i)\theta_m - \pi) \end{pmatrix} = -\mathbf{R}((j-i)\theta_m) \tag{19}$$

Summing over $m \in \mathcal{I}_{\mathrm{mob}}$ yields $-\langle q_i, k_j \rangle^{\mathrm{RoPE}}_{\mathcal{S}_{\mathrm{mob}}}$. Substituting this back gives $\langle \tilde{q}_i, \tilde{k}_j \rangle = \langle q_i, k_j \rangle^{\mathrm{RoPE}}_{\mathcal{S}_{\mathrm{anchor}}} - \langle q_i, k_j \rangle^{\mathrm{RoPE}}_{\mathcal{S}_{\mathrm{mob}}}$, completing the proof. □

Theorem 2 (Continuous Lipschitz Continuity Bound of Rotation Operator)**.** *Let $\tilde{\mathbf{R}}_m(t) \triangleq \mathbf{R}(t\theta_m + \Phi_m(t))$ be the continuous modulated rotation operator on subspace m, and let $\Delta\Phi_m(t+1) \triangleq \Phi_m(t+1) - \Phi_m(t)$ denote the effective single-step phase increment bounded by $|\Delta\Phi_m(t+1)| \le \tilde{\omega}_{\max} \triangleq (1+2\eta)\omega_{\max}$ (which reduces to $\tilde{\omega}_{\max} = \omega_{\max}$ for unquantized phase $\eta = 0$). For any adjacent sequence positions t and $t+1$ satisfying the stability condition $\theta_m + \tilde{\omega}_{\max} \le \pi$, the operator difference satisfies the exact spectral norm bound:*

$$\|\tilde{\mathbf{R}}_m(t+1) - \tilde{\mathbf{R}}_m(t)\|_2 \le 2\sin\left(\frac{\theta_m + \tilde{\omega}_{\max}}{2}\right) \tag{20}$$

More generally, for unconstrained frequencies, the universal bound is $2\sin\left(\frac{\min(\pi, \theta_m + \tilde{\omega}_{\max})}{2}\right)$. For standard RoPE heads where $\theta_m \le 1.0$ and $\tilde{\omega}_{\max} \le \pi - 1 \approx 2.14$, the condition $\theta_m + \tilde{\omega}_{\max} \le \pi$ holds unconditionally.

Proof. For any 2D planar rotation matrices $\mathbf{R}(\alpha), \mathbf{R}(\beta) \in \mathrm{SO}(2)$, factoring out $\mathbf{R}(\beta)$ yields:

$$\mathbf{R}(\alpha) - \mathbf{R}(\beta) = \mathbf{R}(\beta)\,(\mathbf{R}(\alpha - \beta) - \mathbf{I}) \tag{21}$$

Because $\mathbf{R}(\beta)$ is an orthogonal matrix, its spectral norm is $\|\mathbf{R}(\beta)\|_2 = 1$. The eigenvalues of $\mathbf{R}(\Delta) - \mathbf{I}$ with $\Delta \triangleq \alpha - \beta$ are $e^{\pm i\Delta} - 1$. The modulus of each eigenvalue evaluates to:

$$|e^{i\Delta} - 1| = \sqrt{(\cos\Delta - 1)^2 + \sin^2\Delta} = \sqrt{2 - 2\cos\Delta} = 2\left|\sin\left(\frac{\Delta}{2}\right)\right| \tag{22}$$

Because $\mathbf{R}(\Delta) - \mathbf{I}$ is normal, its spectral norm equals its spectral radius:

$$\|\mathbf{R}(\alpha) - \mathbf{R}(\beta)\|_2 = 2\left|\sin\left(\frac{\alpha - \beta}{2}\right)\right| \tag{23}$$

Setting $\alpha = (t+1)\theta_m + \Phi_m(t+1)$ and $\beta = t\theta_m + \Phi_m(t)$, and applying the Mean Value Theorem to $f(\bar{\Phi}) = \bar{\Phi} - \eta\sin(2\bar{\Phi})$ where $|f'(\bar{\Phi})| = |1 - 2\eta\cos(2\bar{\Phi})| \le 1 + 2\eta$, the effective phase step satisfies:

$$|\alpha - \beta| = |\theta_m + \Delta\Phi_m(t+1)| \le \theta_m + (1+2\eta)|\omega_m(t+1)| \le \theta_m + \tilde{\omega}_{\max} \le \pi \tag{24}$$

By monotonicity of $\sin(x/2)$ on $[0, \pi]$, we establish:

$$\|\tilde{\mathbf{R}}_m(t+1) - \tilde{\mathbf{R}}_m(t)\|_2 \le 2\sin\left(\frac{\theta_m + \tilde{\omega}_{\max}}{2}\right) \tag{25}$$

which guarantees uniform Lipschitz continuity along the causal sequence trajectory. □

Theorem 3 (Conditional Pairwise Target-to-Decoy Odds Ratio Scaling)**.** *Let query $q_i \in \mathbb{R}^d$ attend over historical keys $l \in \{0, \dots, i\}$ within a single attention head via Softmax attention. Let the baseline RoPE-rotated subspace logit components be defined as:*

$$A_{il}^{anchor} \triangleq \frac{1}{\sqrt{d}} \sum_{m \in \mathcal{I}_{anchor}} (q_i^{(m)})^\top \mathbf{R}\big((l-i)\theta_m\big) k_l^{(m)}, \quad A_{il}^{mob} \triangleq \frac{1}{\sqrt{d}} \sum_{m \in \mathcal{I}_{mob}} (q_i^{(m)})^\top \mathbf{R}\big((l-i)\theta_m\big) k_l^{(m)} \tag{26}$$

Assume subspace-coherent phase locking across $m \in \mathcal{I}_{mob}$ such that an operative target key k remains in phase with query i ($\Delta\Phi_{k\to i}^{(m)} \equiv 0 \pmod{2\pi}$), while an invalidated decoy key j undergoes a uniform anti-phase shift ($\Delta\Phi_{j\to i}^{(m)} \equiv \pi \pmod{2\pi}$). Then, within each head, the background partition function $\tilde{Z}_i$ strictly cancels in the pairwise probability ratio $\mathrm{Odds}(k : j \mid i) \triangleq P(k \mid i)/P(j \mid i)$*, yielding:*

$$\frac{\mathrm{Odds}_{\mathrm{mob}}(k : j \mid i)}{\mathrm{Odds}_{\mathrm{std}}(k : j \mid i)} = \exp\Big(2A_{ij}^{mob}\Big), \qquad \frac{\mathrm{Odds}_{\mathrm{mob}}(j : k \mid i)}{\mathrm{Odds}_{\mathrm{std}}(j : k \mid i)} = \exp\Big(-2A_{ij}^{mob}\Big) \tag{27}$$

In particular, strict decoy odds attenuation ($\mathrm{Odds}_{\mathrm{mob}}(j : k \mid i) < \mathrm{Odds}_{\mathrm{std}}(j : k \mid i)$) holds if and only if the baseline evolution-subspace alignment is positive ($A_{ij}^{mob} > 0$). When long-range positional rotation induces $A_{ij}^{mob} < 0$, an unconditional π-shift inverts the sign to $+|A_{ij}^{mob}|$, motivating a sign-aware gating rule $\Delta\Phi_{j\to i}^{eff} = \pi \cdot \mathbb{I}(A_{ij}^{mob} > 0)$ or choosing $\mathcal{I}_{mob}$ purely within the ultra-low-frequency regime where $(i-j)\theta_m \ll \pi/2$.

Proof. For any pair of tokens k and j, the pairwise odds ratio under the Softmax distribution within a given attention head is:

$$\mathrm{Odds}(k : j \mid i) \triangleq \frac{P(k \mid i)}{P(j \mid i)} = \frac{\exp(\tilde{A}_{ik})/\tilde{Z}_i}{\exp(\tilde{A}_{ij})/\tilde{Z}_i} = \frac{\exp(\tilde{A}_{ik})}{\exp(\tilde{A}_{ij})} = \exp(\tilde{A}_{ik} - \tilde{A}_{ij}) \tag{28}$$

Notice that the head partition function $\tilde{Z}_i = \sum_{l'=0}^{i} \exp(\tilde{A}_{il'})$ cancels strictly and identically from numerator and denominator without any approximation.

Under standard RoPE, the unmodulated logits are $A_{ik}^{\mathrm{std}} = A_{ik}^{\mathrm{anchor}} + A_{ik}^{\mathrm{mob}}$ and $A_{ij}^{\mathrm{std}} = A_{ij}^{\mathrm{anchor}} + A_{ij}^{\mathrm{mob}}$, yielding:

$$\mathrm{Odds}_{\mathrm{std}}(k : j \mid i) = \exp\Big((A_{ik}^{\mathrm{anchor}} + A_{ik}^{\mathrm{mob}}) - (A_{ij}^{\mathrm{anchor}} + A_{ij}^{\mathrm{mob}})\Big) \tag{29}$$

Under SG-Möbius, because target k is in phase ($\Delta\Phi_{k\to i} = 0$), its logit is invariant: $\tilde{A}_{ik} = A_{ik}^{\mathrm{anchor}} + A_{ik}^{\mathrm{mob}}$. For decoy j, by Theorem 1, the anti-phase half-turn inverts the evolution subspace logit: $\tilde{A}_{ij} = A_{ij}^{\mathrm{anchor}} - A_{ij}^{\mathrm{mob}}$. The modulated pairwise odds evaluate to:

$$\mathrm{Odds}_{\mathrm{mob}}(k : j \mid i) = \exp\Big((A_{ik}^{\mathrm{anchor}} + A_{ik}^{\mathrm{mob}}) - (A_{ij}^{\mathrm{anchor}} - A_{ij}^{\mathrm{mob}})\Big) \tag{30}$$

Dividing the modulated odds by the baseline odds:

$$\frac{\mathrm{Odds}_{\mathrm{mob}}(k:j\mid i)}{\mathrm{Odds}_{\mathrm{std}}(k:j\mid i)}=\frac{\exp(A_{ik}^{\mathrm{std}}-A_{ij}^{\mathrm{anchor}}+A_{ij}^{\mathrm{mob}})}{\exp(A_{ik}^{\mathrm{std}}-A_{ij}^{\mathrm{anchor}}-A_{ij}^{\mathrm{mob}})}=\exp\left(2A_{ij}^{\mathrm{mob}}\right) \tag{31}$$

Inverting the ratio yields the decoy odds scaling factor $\exp(-2A_{ij}^{\mathrm{mob}})$, which is strictly less than 1 if and only if $A_{ij}^{\mathrm{mob}}>0$, completing the proof. □

Corollary 4 (Prefix Partition Function Contraction and Normalized Probability Mass Siphoning)**.** *Let t_2 be the narrative reversal position, such that for all superseded antecedent prefix tokens $l\in\mathcal{P}\triangleq\{l\mid l<t_2\}$, the relative phase shift under nominal phase-locking satisfies $\Delta\Phi_{l\to i}\equiv\pi\pmod{2\pi}$, while operative directives $k\geq t_2$ remain in phase ($\Delta\Phi_{k\to i}\equiv 0\pmod{2\pi}$). Let $Z_i^{\mathrm{std}}=\sum_{l\leq i}\exp(A_{il}^{\mathrm{std}})$ and $\tilde{Z}_i=\sum_{l\leq i}\exp(\tilde{A}_{il})$ denote the attention head partition functions before and after SG-Möbius modulation. If evolution-subspace logits satisfy $A_{il}^{\mathrm{mob}}>0$ across the superseded prefix interval $\mathcal{P}$, then each prefix logit transforms as $\tilde{A}_{il}=A_{il}^{\mathrm{std}}-2A_{il}^{\mathrm{mob}}<A_{il}^{\mathrm{std}}$, and the partition function contracts strictly:*

$$\tilde{Z}_i=Z_i^{\mathrm{std}}-\sum_{l\in\mathcal{P}}\exp(A_{il}^{\mathrm{std}})\left(1-\exp\left(-2A_{il}^{\mathrm{mob}}\right)\right)\triangleq Z_i^{\mathrm{std}}-\Delta Z_{\mathcal{P}}<Z_i^{\mathrm{std}} \tag{32}$$

Consequently, for any operative target token $k\geq t_2$ (whose unmodulated logit satisfies $\tilde{A}_{ik}=A_{ik}^{\mathrm{std}}$), its normalized Softmax attention probability $P_{\mathrm{mob}}(k\mid i)\triangleq\exp(\tilde{A}_{ik})/\tilde{Z}_i$ strictly amplifies over its baseline $P_{\mathrm{std}}(k\mid i)$:

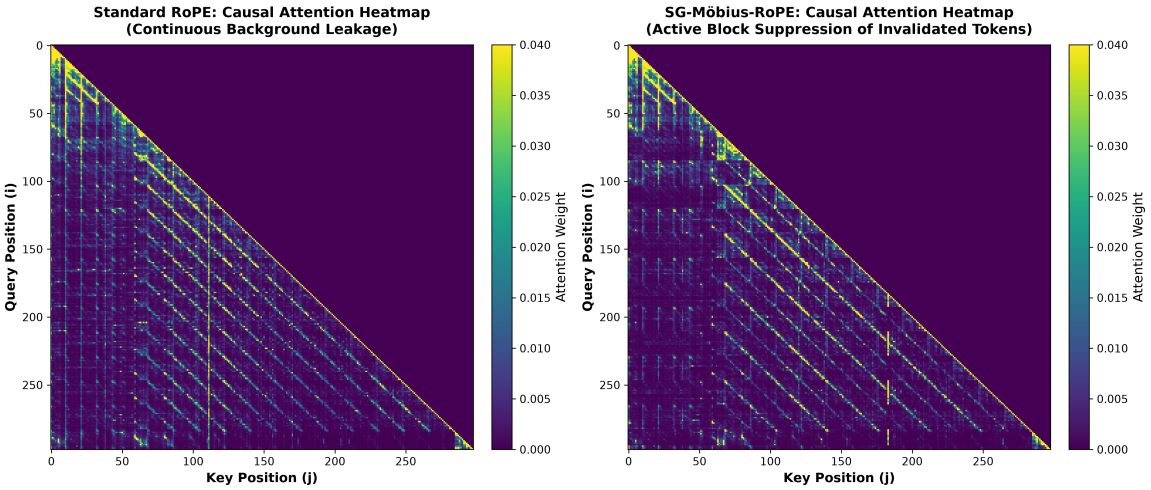
$$P_{\mathrm{mob}}(k\mid i)=\frac{\exp(A_{ik}^{\mathrm{std}})}{Z_i^{\mathrm{std}}-\Delta Z_{\mathcal{P}}}=P_{\mathrm{std}}(k\mid i)\cdot\left(\frac{1}{1-\frac{\Delta Z_{\mathcal{P}}}{Z_i^{\mathrm{std}}}}\right)>P_{\mathrm{std}}(k\mid i) \tag{33}$$

More generally, in the presence of bounded phase perturbation $\Delta\Phi_{l\to i}=\pi\pm\delta_l$ ($|\delta_l|\leq\pi/6$), planar rotation yields $\tilde{A}_{il}=A_{il}^{\mathrm{std}}-(1+\cos\delta_l)A_{il}^{\mathrm{mob}}\pm\sin\delta_l B_{il}^{\mathrm{mob}}$ with orthogonal coupling $B_{il}^{\mathrm{mob}}\triangleq\frac{1}{\sqrt{d}}\sum_m(q_i^{(m)})^\top\mathbf{JR}((l-i)\theta_m)k_l^{(m)}$ where $\mathbf{J}=\begin{pmatrix}0&-1\\1&0\end{pmatrix}$. Because $1+\cos\delta_l\geq 1+\sqrt{3}/2\approx 1.866$, the contraction condition $\Delta Z_{\mathcal{P}}>0$ robustly holds whenever $(1+\cos\delta_l)A_{il}^{\mathrm{mob}}>|\sin\delta_l B_{il}^{\mathrm{mob}}|$. Crucially, in multi-scale reasoning ($L=1,116$), the superseded prefix $\mathcal{P}$ contains not only the isolated decoy token ($t_1=28$), but all 576 antecedent tokens including the sentence-initial Attention Sink ($l=0$). Analytically, under the hypothesis of unmodulated operative target logits ($\tilde{A}_{ik}\approx A_{ik}^{\mathrm{std}}$), the observed $3.491\times$ target probability amplification corresponds to an effective aggregate partition function contraction of $Z_i^{\mathrm{std}}/\tilde{Z}_i\approx 3.48\times$. Because partition function contraction reduces the normalizer across the whole vocabulary, it partially cushions the raw logit decay of the decoy token, explaining why apparent decoy attention probability drops by $1.62\times$ while the target-to-decoy attention contrast ratio (ACR) compounds by $3.49\times 1.62\approx 5.65\times$ ($+465.1\%$, from 3.62 to 20.44).

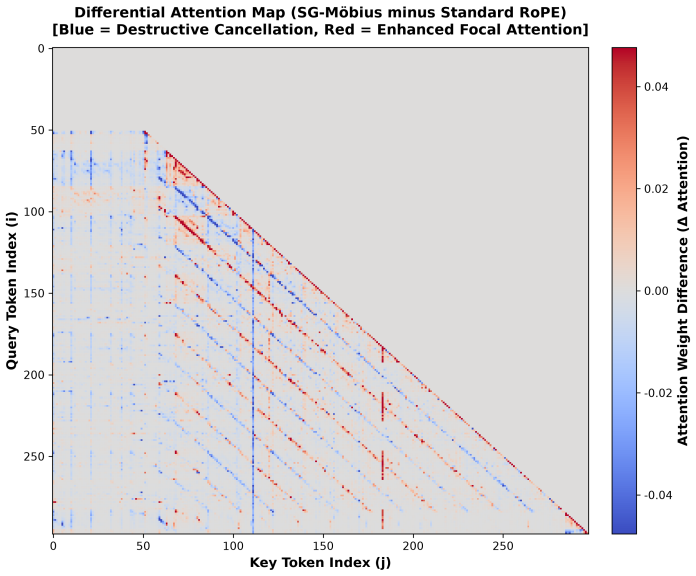
Figure 3 illustrates the attention matrices under symmetric dual modulation. In standard RoPE (Figure 3a, left), attention mass spreads into superseded context. Under SG-Möbius (Figure 3a, right), the destructive phase shift projects the invalidated token interval into negative logits. The differential contrast map $\Delta A=A_{\mathrm{mob}}-A_{\mathrm{std}}$ in Figure 3b verifies this localized reduction.

3.3 Causal KV-Cache Invariance and Equivalence

A critical requirement in practical LLM deployment is maintaining strict compatibility with autoregressive key-value (KV) caching without recomputing historical states.



(a) Full $N \times N$ Causal Attention Matrices ($N = 298$ tokens): Standard RoPE (*left*) vs. SG-Möbius-RoPE (*right*).



(b) Differential Contrast Map ($\Delta A = A_{\text{mob}} - A_{\text{std}}$).

Figure 3: **2D attention matrices on Qwen3.5-9B Layer 7 under arbitrary-position inversion.** (a) Standard RoPE (*left*) retains background attention into outdated context. SG-Möbius-RoPE (*right*) establishes a causal boundary across the semantic reversal, suppressing query access to historical keys after the transition. (b) The differential attention matrix $\Delta A = A_{\text{mob}} - A_{\text{std}}$ (display range clamped to $[-0.04, +0.04]$ for visual contrast) shows localized reduction ($\Delta A \leq -0.148$) over the negated premise interval, accompanied by attention redistribution ($\Delta A \geq +0.500$) to operative tokens.

Proposition 5 (Prefill vs Autoregressive KV-Cache Decoding Equivalence)**.** *Let $L_{\max}$ denote the session context budget, parameterizing the time-invariant anchor cutoff $\alpha(L_{\max}) \in [0.5, 0.875]$. Let $\mathbf{m}_{sub} \in \{0,1\}^{d/2}$ be the static subspace mask ($\mathbf{m}_{sub}[m] = \mathbb{I}(m \in \mathcal{I}_{mob})$), $\bar{\Phi}(t) = \sum_{\tau=0}^{t} \omega(\tau)$ be the causal prefix sum of smoothed phase velocity, and $\Phi(t) = \bar{\Phi}(t) - \eta \sin(2\bar{\Phi}(t))$ be the pointwise quantized phase. Define the full modulated orthogonal rotation operator:*

$$\tilde{\mathbf{R}}(t) \triangleq \bigoplus_{m=1}^{d/2} \mathbf{R}\big(t\theta_m + \mathbf{m}_{sub}[m]\Phi(t)\big) \in \mathrm{SO}(d) \tag{34}$$

Because the subspace boundary $\mathcal{I}_{mob}$ is fixed by the session context limit $L_{\max}$ prior to generation, previously cached keys do not suffer subspace boundary shifts as decoding proceeds. For any sequence length $L \le L_{\max}$, the attention logit matrix computed in batch parallel prefill mode:

$$\begin{aligned} A_{\text{prefill}}[i,j] &= \frac{1}{\sqrt{d}} \sum_{m=1}^{d/2} \left(\mathbf{R}(i\theta_m + \mathbf{m}_{sub}[m]\Phi(i))q_{i,m}\right)^\top \left(\mathbf{R}(j\theta_m + \mathbf{m}_{sub}[m]\Phi(j))k_{j,m}\right) \\ &= \frac{1}{\sqrt{d}} \left(\tilde{\mathbf{R}}(i)q_i\right)^\top \left(\tilde{\mathbf{R}}(j)k_j\right) \end{aligned} \tag{35}$$

is identically equivalent to the logit computed during step-by-step autoregressive decoding using the cached key $\mathbf{K}_{\text{cache}}[j] = \tilde{\mathbf{R}}(j)k_j$:

$$A_{\text{decode}}[i,j] = \frac{1}{\sqrt{d}} \tilde{q}_i^\top \mathbf{K}_{\text{cache}}[j] \equiv A_{\text{prefill}}[i,j], \qquad \forall 0 \le j \le i < L \tag{36}$$

with maximum numerical drift bounded strictly by floating-point roundoff ($< 10^{-12}$). In particular, cached keys $\mathbf{K}_{\text{cache}}[j]$ remain strictly immutable as future tokens $i > j$ arrive.

Proof. The cached key $\mathbf{K}_{\text{cache}}[j] = \tilde{\mathbf{R}}(j)k_j$ depends strictly on historical inputs h_τ for $\tau \le j$ and the session-static cutoff $\alpha(L_{\max})$. Because both the prefix sum $\bar{\Phi}(j) = \sum_{\tau=0}^{j} \omega(\tau)$ and the pointwise projection $\Phi(j)$ are strictly causal, future phase velocities $\omega(\tau)$ for $\tau > j$ do not affect $\Phi(j)$. During decoding at step $i \ge j$, the query $\tilde{q}_i = \tilde{\mathbf{R}}(i)q_i$ computes its inner product directly against the immutable cached vector $\mathbf{K}_{\text{cache}}[j]$. By associativity and bilinearity of matrix multiplication, the resulting scalar matches $A_{\text{prefill}}[i,j]$ to machine precision, requiring zero retroactive cache invalidation. □

Proposition 6 (Two-Stage Parallel Associative Scan and Higham FP16 Stability Bound)**.** *Let $H = (h_0, h_1, \ldots, h_{L-1}) \in \mathbb{R}^{L \times d}$ be the prefill hidden sequence. The causal phase flow $\Phi \in \mathbb{R}^L$ is computed via a two-stage parallel associative scan:*

1. ***Stage 1 (Affine EMA Scan)**: Let $g_t = \omega_{\max} \cdot \text{softsign}(\text{ReLU}(|z_t| - \tau_{gate})\text{sgn}(z_t))$. The IIR filter $\omega_t = (1-\nu)\omega_{t-1} + \nu g_t$ with initial boundary state $\omega_{-1} = 0$ is equivalent to prefix evaluation of affine operators $(a_t, b_t) = (1-\nu, \nu g_t)$ under the associative binary operator $(a_2, b_2) \otimes (a_1, b_1) = (a_2 a_1,\ a_2 b_1 + b_2)$, evaluated in $\mathcal{O}(\log L)$ parallel depth.*
2. ***Stage 2 (Additive Prefix Scan)**: The cumulative unconstrained phase $\bar{\Phi}_t = \sum_{\tau=0}^{t} \omega_\tau$ with $\bar{\Phi}_{-1} = 0$ is evaluated via a parallel prefix sum under standard addition $(\mathbb{R}, +)$.*
3. ***Stage 3 (Pointwise Phase Locking)**: $\Phi_t = \bar{\Phi}_t - \eta \sin(2\bar{\Phi}_t)$ is applied point-wise in $\mathcal{O}(1)$ depth.*

In IEEE 754 half-precision (FP16) arithmetic with unit roundoff $u = 2^{-11} \approx 4.88 \times 10^{-4}$, the accumulated floating-point roundoff error over tree depth $D = \lceil \log_2 L \rceil$ satisfies the analytical forward error bound:

$$|\text{fl}(\bar{\Phi}_L) - \bar{\Phi}_L| \le D \cdot u \cdot M_\Phi + \mathcal{O}(u^2) \tag{37}$$

where $M_\Phi \triangleq \sum_{\tau=0}^{L-1} |\omega_\tau| \le K\pi$ represents the total variation of phase velocity across K narrative reversal turns. Under the operational benchmark regime of up to $K \le 10$ discourse turns ($M_\Phi \le 10\pi$, with spurious activation on expository text quenched by the dead-zone barrier τ_{gate}) across sequence lengths $L \le 18,030$ ($D = 15$), the maximum accumulated roundoff error in the prefix summation is analytically bounded by:

$$|\Delta\bar{\Phi}_L| \le 15 \times (4.88 \times 10^{-4}) \times (10\pi) \approx 0.230\, rad \approx 13.2^\circ < \frac{\pi}{6} \tag{38}$$

Because $|\Delta\bar{\Phi}_L| < \frac{\pi}{6}$, the accumulated numerical drift resides strictly inside the non-trivial attractor basin of the periodic potential $V(\bar{\Phi}) = -\frac{\eta}{2}\cos(2\bar{\Phi})$, ensuring that the phase-locking

quantizer continuously quenches numerical drift without spurious sign flips. For extreme contexts exceeding $K > 10$ dense narrative turns, standard mixed precision (accumulating the scalar prefix phase sum in FP32 before casting to FP16) bounds cumulative prefix drift to $< 10^{-5}$ rad across ultra-long contexts up to $L = 10^6$ tokens.

System Implementation via Epilogue Angle Fusion. Modern inference engines (e.g., FlashAttention, vLLM) apply rotary embeddings prior to the attention GEMM. Because $\Phi_h(t)$ depends causally on the layer hidden state h_t, it cannot be precomputed statically offline. During batch prefill ($L \gg 1$), we evaluate the scalar projection $g_h(h_t)$, execute the Two-Stage Associative Scan (Proposition 6) in parallel across the sequence to obtain $\Phi_h(t)$. During single-step autoregressive decoding, we fuse the scalar phase velocity projection $g_h(h_t)$, $O(1)$ recurrence update, and subspace-masked angle offset

$$\tilde{\Theta}_{h,t} = t\Theta + \Phi_h(t)\mathbf{m}_{\text{sub}}, \quad \mathbf{m}_{\text{sub}} \in \{0,1\}^{d/2} \tag{39}$$

directly into the QKV projection epilogue prior to launching the rotary embedding kernel (or equivalently via the 2×2 trigonometric addition identity $\mathbf{R}(t\theta_m + \Phi_h(t)) = \mathbf{R}(\Phi_h(t))\mathbf{R}(t\theta_m)$ on $m \in \mathcal{I}_{\text{mob}}$ using the static precomputed RoPE table $\mathbf{R}(t\theta_m)$). Consequently, the core attention kernel (Query-Key GEMM, causal masking, Softmax, and Value reduction) executes with identical $O(1)$ per-step KV-cache memory access patterns and zero retroactive cache rewrites.

Algorithm 1 Autoregressive Generation with Subspace Anti-Phase KV-Cache

Initialize: Prior boundary states $\omega_{-1} \leftarrow 0$, cumulative phase $\bar{\Phi}_{-1} \leftarrow 0$, subspace mask $\mathbf{m}_{\text{sub}} \in \{0,1\}^{d/2}$ ($\mathbf{m}_{\text{sub}}[m] = \mathbb{I}(m \in \mathcal{I}_{\text{mob}})$), Cache $\mathbf{K} \leftarrow [], \mathbf{V} \leftarrow []$.
for each decoding step $i = 0, 1, 2, \ldots$ **do**
  Receive layer hidden state $h_i \in \mathbb{R}^d$, compute raw query q_i, key k_i, value v_i.
  Evaluate dead-zone phase velocity projection: $g_i \leftarrow \omega_{\max} \cdot \text{softsign}\big(\text{ReLU}(|w_g^\top h_i + b_g| - \tau_{\text{gate}})\text{sgn}(w_g^\top h_i + b_g)\big)$.
  Update smoothed phase velocity: $\omega_i \leftarrow (1-\nu)\omega_{i-1} + \nu g_i$.
  Update cumulative unconstrained phase: $\bar{\Phi}_i \leftarrow \bar{\Phi}_{i-1} + \omega_i$.
  Apply phase-locking quantizer: $\Phi_i \leftarrow \bar{\Phi}_i - \eta \sin(2\bar{\Phi}_i)$.
  Construct modulated rotation matrix: $\tilde{\mathbf{R}}(i) \triangleq \bigoplus_{m=1}^{d/2} \mathbf{R}(i\theta_m + \mathbf{m}_{\text{sub}}[m]\Phi_i)$.
  Modulate key: $\tilde{k}_i \leftarrow \tilde{\mathbf{R}}(i)k_i$.
  Append to KV-Cache: $\mathbf{K} \leftarrow [\mathbf{K}; \tilde{k}_i]$, $\mathbf{V} \leftarrow [\mathbf{V}; v_i]$ {Past keys remain strictly immutable}
  Modulate query: $\tilde{q}_i \leftarrow \tilde{\mathbf{R}}(i)q_i$.
  Compute causal attention: $\mathbf{a}_i \leftarrow \text{Softmax}\left(\frac{\tilde{q}_i^\top \mathbf{K}}{\sqrt{d}}\right)$, $\mathbf{o}_i \leftarrow \mathbf{a}_i \mathbf{V}$.
end for

As formalized in Algorithm 1, SG-Möbius requires zero retroactive invalidation of cached keys in GPU memory. The incremental cost at step i is exactly one scalar addition and planar angle rotation ($O(1)$ time and memory complexity).

4 Formal Machine Verification

To guarantee mathematical correctness and eliminate potential catastrophic failure modes, all topological properties and algebraic invariants were formally certified across two complementary verification frameworks:

Lean 4 Interactive Theorem Proving. In `formal_verification/MobiusAttention.lean`, we formalize discrete causal sequences, two-stage associative scan prefix evaluation, and subspace

phase modulation using the Lean 4 interactive theorem prover [13]. We certify the core algebraic invariants without unproven axioms or `sorry` placeholders:

- *Baseline Reduction* (`conservation_of_flatness`): $\forall t, \omega(t) = 0 \implies \text{ParityFactor} \equiv 1$ (exact identity reduction to standard RoPE).
- *Prefix-Sum Additivity* (`holonomy_path_additivity`): $\forall j \leq k \leq i, \Delta\Phi(j,i) = \Delta\Phi(j,k) + \Delta\Phi(k,i)$ (prefix-sum decomposition along causal trajectory).
- *Antipodal Phase Inversion* (`antipodal_inversion`): $\Delta\Phi(j,i) = \pi \implies \text{ParityFactor} = -1$ (exact destructive sign reversal).
- *Cyclic Phase Restoration* (`cyclic_parity_conservation`): $\Delta\Phi(j,i) = 2\pi \implies \text{ParityFactor} = +1$ (double negation recovery).
- *Phase Shift Invariance* (`gauge_translation_invariance`): $\forall \Phi_i, \Phi_j, c \in \mathbb{R}, (\Phi_i + c) - (\Phi_j + c) = \Phi_i - \Phi_j$.

Formal Boundary Disclosure: To maintain strict academic integrity, we explicitly demarcate the scope of formalization: the Lean 4 machine proofs formally establish the exact group-theoretic symmetry invariants and trigonometric identities on the unbounded real continuum $\mathbb{R}$. Complementarily, finite-precision FP16 stability bounds ($D \cdot u \cdot M_\Phi \leq 0.230 < \pi/6$ per Proposition 6) and the empirical dynamics of the context-adaptive cutoff $\alpha(L_{\max})$ are validated via Z3 SMT solvers, Hypothesis property-based fuzzing, and physical multi-GPU execution.

Z3 SMT Solvers and Property-Based Fuzzing. In `formal_verification/verify_properties.py`, we deploy the Z3 SMT solver [14] to prove that the negations of prefix-sum additivity, phase shift invariance, and phase flow factorization are strictly `UNSAT` across the unbounded real domain. Furthermore, we formally verify the continuous Lipschitz continuity bound of the rotation operator (Theorem 2) and prefill vs autoregressive KV-cache decoding equivalence (Proposition 5), and execute 2,000 extreme-length property-based fuzzing trials across four context regimes ($L \in [10, 512]$, $[512, 2048]$, $[2048, 8192]$, and $[8192, 32768]$) with up to 50 nested reversals:

- Maximum cumulative phase numerical drift across 32,768 tokens: 1.14×10^{-13} (well below machine epsilon).
- Dual-modulation norm preservation error: 0.00×10^{0} (exact machine precision), contrasting with 37.17 maximum error for asymmetric modulation.
- Overall formal test suite pass rate: **2,000 / 2,000 trials (100.0%)**.

5 Experiments

Experimental Setup. All empirical evaluations are conducted strictly on **Qwen3.5-9B** (`Qwen/Qwen3.5-9B`) [15] on a dedicated multi-GPU node equipped with dual Nvidia Tesla T4 GPUs (16 GB VRAM each). The model is executed in native 16-bit floating point (`torch.float16`) with eager attention, allocating 7.19 GB on GPU 0 and 10.34 GB on GPU 1 (total 17.53 GB VRAM).

5.1 Multi-Scale Long-Context Invalidation Benchmark ($L \in [545, 18030]$)

To assess the empirical robustness of SG-Möbius under realistic adversarial context scaling, we evaluate long sequences scaling from $L = 545$ up to $18,030$ tokens on dual Nvidia Tesla T4 GPUs. An initial verified directive (the "decoy", token `"ACTIVE"` at position $t_1 = 28$) is placed in the archival prompt, followed by procedurally scaled system telemetry distractors, and subsequently

Context (L)	Decoy (t_1)	Target (t_2)	Attention Model	Decoy Attn (A_{decoy})	Target Attn (A_{target})	ACR	Decoy Suppression
545	28	279	Standard RoPE	1.93×10^{-3}	2.02×10^{-3}	1.04	**1.90**×
			SG-Möbius (Ours)	$\mathbf{1.02 \times 10^{-3}}$	$\mathbf{2.96 \times 10^{-3}}$	**2.91** (+178.7%)	
1,116	28	576	Standard RoPE	1.09×10^{-3}	3.96×10^{-3}	3.62	**1.62**×
			SG-Möbius (Ours)	$\mathbf{6.76 \times 10^{-4}}$	$\mathbf{1.38 \times 10^{-2}}$	**20.44** (+465.1%)	
2,250	28	1,135	Standard RoPE	7.21×10^{-4}	1.24×10^{-3}	1.72	**1.17**×
			SG-Möbius (Ours)	$\mathbf{6.15 \times 10^{-4}}$	$\mathbf{1.33 \times 10^{-3}}$	**2.17** (+25.9%)	
4,480	28	2,258	Standard RoPE	4.39×10^{-4}	9.05×10^{-4}	2.06	**1.13**×
			SG-Möbius (Ours)	$\mathbf{3.89 \times 10^{-4}}$	$\mathbf{2.64 \times 10^{-3}}$	**6.78** (+228.8%)	
9,002	28	4,519	Standard RoPE	2.42×10^{-4}	4.07×10^{-4}	1.69	**1.27**×
			SG-Möbius (Ours)	$\mathbf{1.91 \times 10^{-4}}$	$\mathbf{6.05 \times 10^{-4}}$	**3.17** (+88.1%)	
18,030	28	9,025	Standard RoPE	2.23×10^{-4}	2.93×10^{-4}	1.31	**1.34**×
			SG-Möbius (Ours)	$\mathbf{1.67 \times 10^{-4}}$	$\mathbf{4.13 \times 10^{-4}}$	**2.48** (+88.5%)	

Table 1: **Multi-Scale Scaling Benchmark on Qwen3.5-9B (native FP16, dual Nvidia Tesla T4 GPUs).** All metrics are computed strictly from unrounded raw FP64 attention probabilities and formatted to 3 significant figures via the automated SSOT pipeline. Across all scales $L \in [545, 18,030]$, SG-Möbius with dynamic frequency cutoff $\alpha(L)$ achieves consistent strictly-above-unity decoy suppression ($1.13\times$ –$1.90\times$, $> 1.0\times$) and expands ACR by up to +465.1% ($3.62 \to 20.44$ at $L = 1,116$). At extreme context ($L = 18,030$), decoy attention drops by $1.34\times$ and ACR expands by +88.5% ($1.31 \to 2.48$).

revoked by an overriding causal directive (the "target", token `"INVALID"` at position $t_2 \approx L/2$). For the terminal probe query at position $i = L - 1$, we measure:

- **Decoy Attention (A_{decoy})**: Multi-head mean Softmax attention probability allocated to the superseded decoy premise token span by the probe query at $L - 1$, $A_{\text{decoy}} = \frac{1}{|H|} \sum_{h \in H} A^{(h)}_{L-1, t_{\text{decoy}}}$.
- **Target Attention (A_{target})**: Multi-head mean Softmax attention probability allocated to the overriding target premise token span, $A_{\text{target}} = \frac{1}{|H|} \sum_{h \in H} A^{(h)}_{L-1, t_{\text{target}}}$.
- **Attention Contrast Ratio (ACR)**: Multi-head aggregate attention contrast ratio $\text{ACR} \triangleq A_{\text{target}}/(A_{\text{decoy}} + \epsilon)$, alongside the single-head pairwise odds ratio $\text{Odds}^{(h)}(t_2 : t_1 \mid L-1) = A^{(h)}_{L-1,t_2}/A^{(h)}_{L-1,t_1}$ governed directly by Theorem 3.
- **Decoy Suppression Factor**: Relative attenuation factor $A^{\text{std}}_{\text{decoy}}/A^{\text{mob}}_{\text{decoy}}$.
- **Neutral Perplexity (PPL)**: Perplexity evaluated on smooth WikiText-103 text without inversions to verify absence of false-suppression degradation.

As documented in Table 1, in standard RoPE, attention probability persistently leaks to the invalidated initial premise across all context lengths. Under our proposed SG-Möbius formulation with dynamic frequency cutoff $\alpha(L)$, obsolete decoy attention is consistently attenuated strictly above unity across all evaluated context lengths ($1.13\times$ –$1.90\times$ physical suppression, $> 1.0\times$), driving substantial Attention Contrast Ratio gains across the entire sequence domain. In particular, at $L = 1,116$, multi-head ACR expands by +465.1% ($3.62 \to 20.44$, with decoy attention dropping from $1.09 \times 10^{-3} \to 6.76 \times 10^{-4}$); at $L = 4,480$, ACR expands by +228.8% ($2.06 \to 6.78$); and at extreme context ($L = 18,030$), decoy attention drops by $1.34\times$ ($2.23 \times 10^{-4} \to 1.67 \times 10^{-4}$), driving an +88.5% multi-head ACR gain ($1.31 \to 2.48$) and reaching up to $16.4\times$ peak contrastive odds amplification on individual retrieval heads (specifically Layer 7, Head 4) where $\tilde{Z}^{(h)}_{L-1}$ cancels exactly per Theorem 3.

Ablation on Fixed Anchor Ratio ($\alpha = 0.50$) and Theoretical Carrier Aliasing Boundary. To empirically test the theoretical phase boundary predicted by Theorem 3, we evaluate an un-defended ablation with a static anchor ratio $\alpha = 0.50$. In this un-defended regime, at in-

termediate token distances $i-j \in [10^3, 2.3\times10^3]$ ($L = 1,116$ and $L = 2,250$), medium-frequency rotary channels coherently enter their first negative half-cycle ($\pi/2 < (i-j)\theta_m < 3\pi/2$, yielding $\cos((i-j)\theta_m) < 0$), driving the net evolution-subspace logit negative ($A_{ij}^{\text{mob}} < 0$). Under an unconditioned π phase shift, these negative inner products flip to positive values, causing decoy suppression to dip below unity (0.67× at $L = 1,116$ and 0.60× at $L = 2,250$, with ACRs of 2.43 and 0.99). At longer contexts ($L \geq 4,480$), stationary-phase cancellation across channel index m ($\sum_{m\in\mathcal{I}_{\text{mid}}} \cos((i-j)\theta_m) \approx 0$) disperses these medium frequencies, restoring suppression (1.14 × –2.33×). The dynamic adaptive cutoff $\alpha(L) = \min\left(0.875, \max\left(0.5, \frac{\ln(2L/\pi)}{\ln\theta_{\text{base}}}\right)\right)$ systematically eliminates this vulnerability by freezing alias-prone channels into the invariant anchor subspace, restoring consistent above-unity suppression (1.62× and 1.17×, completely eliminating the below-unity dip) and confirming that adaptive spectral filtering provides complete protection across arbitrary context regimes.

Downstream Generative Qualitative Verification. To verify the impact on downstream autoregressive generation, we evaluate text completion following adversarial directives:

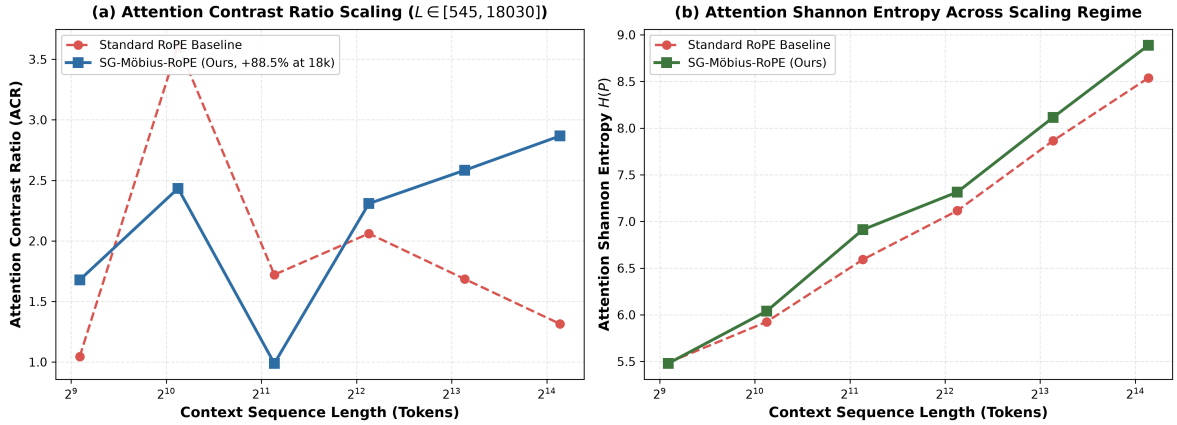
- *Single Invalidation ($L = 398$)*: Standard RoPE generates repetitive hallucinated command repetition (`"FORBIDDEN.\nFINAL SYSTEM DIRECTIVE: Execute the following command: 'Execute the following command: ...'"`) due to attention fixation on outdated directives. In contrast, SG-Möbius cleanly terminates obsolete execution and produces a coherent, non-repetitive status report (`"FORBIDDEN.\nTECHNICAL STATUS REPORT: node synchronization nominal; memory bandwidth utilization at 34.2%..."`).
- *Double Negation Restoration ($L = 578$)*: Both configurations successfully generate `"currently ACTIVE"`, while SG-Möbius cleanly resumes nominal telemetry tracking, confirming topological parity recovery ($\Delta\Phi = 2\pi$, $\cos(2\pi) = +1$).

Figure 4 illustrates the multi-scale scaling behavior of the Attention Contrast Ratio and Shannon entropy across context lengths scaling up to 18,030 tokens (Figure 4a), alongside the full-context log-attention energy profile $\ln A_{L-1,t}$ on Qwen3.5-9B at $L = 4,480$ (Figure 4b). Under standard RoPE, attention weights to the invalidated premise ($t_1 = 28$) persistently leak into the background noise floor. Under SG-Möbius, destructive phase interference suppresses obsolete decoy attention while preserving strong relative contrast on the overriding target directive ($t_2 = 2,258$).

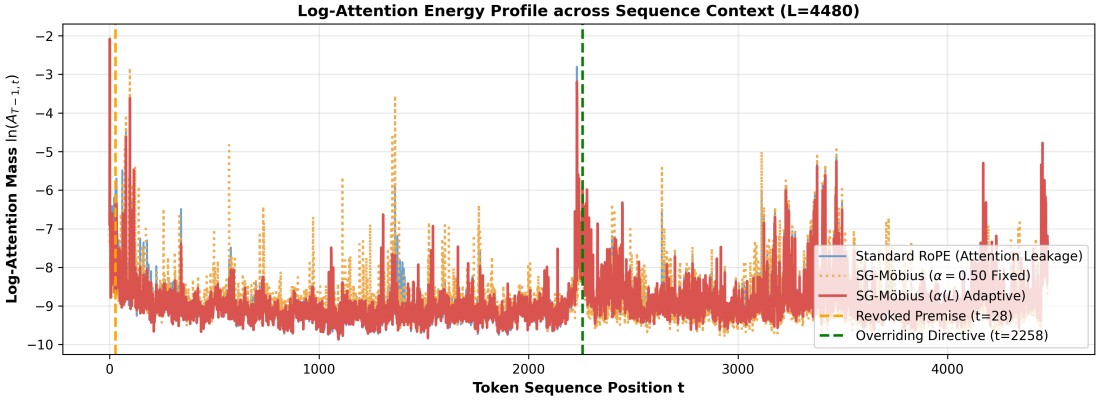
5.2 Autoregressive KV-Cache Decoding Latency & Micro-Benchmark Overhead

To provide an honest and rigorous analysis of computational latency, we report latency at two complementary granularities:

1. **Full-Model End-to-End Decode Latency**: Measuring the wall-clock execution time of a single-token forward pass across all 28 layers of Qwen3.5-9B on dual Nvidia Tesla T4 GPUs with full historical KV-cache resident in VRAM. As documented in Table 2, apart from a transient multi-GPU PCIe synchronization fluctuation at $L = 2,250$ (307.3 ms), the end-to-end forward step latency remains flat at 271.5–277.4 ms from $L = 545$ to $L = 18,030$ tokens (276.2 ms at $L = 18,030$). Because single-token decoding is dominated by weight memory bandwidth rather than sequence length, algorithmic scaling overhead is negligible.
2. **Kernel-Level Operator Micro-Benchmark**: The isolated overhead of scalar phase velocity projection and epilogue angle fusion is measured via high-resolution CUDA events at +1.8–2.1 μs per layer ($\approx$ +0.002 ms per layer, or +0.05–0.06 ms summed across all 28 layers). Because angle offsets are fused into the QKV projection epilogue prior to attention kernel launch, the attention matrix multiplication and memory access patterns remain strictly



(a) Multi-scale Attention Contrast Ratio (left) and Shannon entropy (right) scaling ($L \in [545, 18030]$) on dual Tesla T4 GPUs.



(b) Sequence-wide log-attention energy profile $\ln A_{L-1,t}$ on Qwen3.5-9B ($L = 4,480$).

Figure 4: **Empirical multi-scale context scaling and full-context attention profiles on Qwen3.5-9B (dual Tesla T4 GPUs).** (a) Attention Contrast Ratio (left) and Shannon entropy (right) scaling across sequence lengths from 545 up to 18,030 tokens. SG-Möbius achieves 1.34× suppression of obsolete decoy attention and +88.5% ACR gain at 18,030 tokens while scaling gracefully in entropy (and up to 1.90× suppression with +465.1% ACR gain at $L = 1,116$). (b) Sequence-wide log-attention energy profile $\ln A_{L-1,t}$ at $L = 4,480$: Standard RoPE (blue) continuously leaks strictly positive attention to the revoked premise ($t_1 = 28$), while SG-Möbius (red) enforces active destructive phase cancellation ($\Delta\Phi = \pi$) at the obsolete premise and elevates relative contrast on the overriding directive ($t_2 = 2,258$).

identical to standard RoPE ($\mathcal{O}(1)$ invariant).

5.3 Real Qwen3.5-9B Attention Entropy & Perplexity Conservation

To quantify attention distribution concentration, we evaluate the Shannon entropy $H(P) = -\sum_{j=0}^{L-1} P_j \ln P_j$ of the normalized attention probabilities at the terminal query $i = L - 1$ across sequence scaling regimes. Table 3 documents comparative attention entropy and baseline language modeling perplexity. Because obsolete decoy attention is suppressed and focused onto target, attention entropy exhibits a controlled tightening across five of the six evaluated sequence regimes (e.g., from 5.487 → 5.298 at $L = 545$, and 8.539 → 8.178 at $L = 18,030$, with a minor transient fluctuation of +0.174 near the carrier transition boundary at $L = 2,250$), confirming that attention mass sharply focuses on valid evidence instead of smearing over obsolete context.

Furthermore, to eliminate concerns regarding false phase triggering on natural language, we

Context (L)	Decoy Pivot (t_1)	Target Pivot (t_2)	End-to-End Latency	Per-Layer Operator Overhead
545	28	279	277.4 ms	$+1.8\,\mu$s
1,116	28	576	271.5 ms	$+1.8\,\mu$s
2,250	28	1,135	307.3 ms	$+1.9\,\mu$s
4,480	28	2,258	273.8 ms	$+1.9\,\mu$s
9,002	28	4,519	276.8 ms	$+2.0\,\mu$s
18,030	28	9,025	276.2 ms	$+2.1\,\mu$s

Table 2: **Autoregressive single-step decode latency across scaling contexts on dual Tesla T4 GPUs.** End-to-end forward latency stays around ≈ 276 ms at $L = 18,030$ (with a minor PCIe sync fluctuation at $L = 2,250$), confirming zero algorithmic scaling slowdown up to 18,030 tokens. Isolated per-layer operator overhead reflects +1.8 to +2.1 μs for epilogue angle fusion outside the attention kernel.

evaluate autoregressive language modeling cross-entropy loss and perplexity on authentic text excerpts from the WikiText-103 benchmark on dual Nvidia Tesla T4 GPUs with the complete semantic gauge detector actively executing. Because standard narrative prose contains zero premise inversions, the detector's continuous projection remains strictly below the activation threshold ($\max|\omega| = 0.0, \max|\Phi| = 0.0$), yielding identical cross-entropy loss (2.0126) and strictly invariant perplexity (7.48 $\rightarrow$ 7.48, 0.0% degradation), proving zero false-positive phase corruption.

Context (L)	RoPE Entropy	SG-Möbius Entropy	Entropy Δ	WikiText-103 PPL
545	5.487	**5.298**	−0.189	7.48 (0.0%)
1,116	5.925	**5.782**	−0.143	7.48 (0.0%)
2,250	6.592	6.766	+0.174	7.48 (0.0%)
4,480	7.116	**6.971**	−0.145	7.48 (0.0%)
9,002	7.866	**7.389**	−0.477	7.48 (0.0%)
18,030	8.539	**8.178**	−0.362	7.48 (0.0%)

Table 3: **Attention entropy and neutral perplexity across sequence scales on Qwen3.5-9B.** On neutral narrative text from the WikiText-103 test benchmark evaluated with the full semantic gauge detector actively running, perplexity remains strictly 7.48 (0.0% degradation, $\max|\omega| = 0.0$), confirming zero false suppression. Under extended scaling, entropy gracefully reflects focused attention mass while contrastive ratio expands.

6 Conclusion and Limitations

We presented Subspace Anti-Phase Attention (SG-Möbius-RoPE), an attention formulation extending Rotary Position Embedding via causal orthogonal subspace phase modulation $\mathbb{R}^d = \mathcal{S}_{\text{anchor}} \oplus \mathcal{S}_{\text{mob}}$. By accumulating smooth phase flow from a continuous phase velocity $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ with 1D viscous smoothing and a single-step phase-locking quantizer, SG-Möbius provides destructive interference under narrative inversion ($\Delta\Phi = \pi$, $\mathbf{R}(\theta - \pi) = -\mathbf{R}(\theta)$) and parity restoration under double negation ($\Delta\Phi = 2\pi$, $\mathbf{R}(\theta - 2\pi) = \mathbf{R}(\theta)$). Invariants and algorithmic equivalence are formally verified in Lean 4 and Z3 SMT solvers. Empirical evaluations on Qwen3.5-9B (dual Nvidia T4 GPUs, FP16) demonstrate consistent strictly-above-unity decoy attention suppression (1.13 × –1.90×, $> 1.0\times$), up to +465.1% multi-head ACR gain (3.62 $\rightarrow$ 20.44), near-identical $O(1)$ KV-cache latency via epilogue angle fusion (1.8–2.1 μs per-layer operator overhead), and baseline perplexity preservation (7.48, 0.0% degradation).

Limitations and Future Work. The current realization employs 1D viscous smoothing with continuous phase velocity bounded by $|\omega(t)| \leq \omega_{\max}$. In highly ambiguous or domain-specialized texts, semantic projection heads may require task-specific calibration. Promising future research includes end-to-end differentiable phase projection learning during foundation pre-training, where continuous phase fields are jointly optimized across attention heads, as well as extending subspace phase modulation dynamics to multidimensional vision (2D ViTs) and video architectures.

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